

Isaac Han

✉ Lssac7778@gm.gist.ac.kr isaac7778.github.io [Github](#) [Google Scholar](#)

Research Interests

I'm interested in building an intelligent agent that can continuously learn and adapt to changing environments. My current and future research focus on reinforcement learning, continual learning, and streaming learning.

Previously, I explored wide range of topics such as 3D pose estimation from wearable sensors, StarCraft AI, VR haptic devices, and off-road autonomous driving.

Education

Gwangju Institute of Science and Technology (GIST) 2020 - 2026

M.S/Ph.D student in AI (GPA: 4.33/4.5).

- Advisor: [Kyung-Joong Kim](#).

Bachelor's Degree Examination for Self-Education 2019

B.S in Computer Science (GPA: 3.7/4.3).

- Nationally accredited B.S. awarded by [NILE](#) (Ministry of Education, Korea).

Research Experience

University of Alberta Mar 2026 - Jun 2026

Visiting Researcher.

- Advisor: [Rupam Mahmood](#).
- Improving sample efficiency of deep RL by scaling the replay ratio with recent advances in [streaming RL](#).

Publications

Conferences and Journals

FIRE: Frobenius-Isometry Reinitialization for Balancing the Stability-Plasticity Tradeoff ICLR'26
Oral Presentation

- **Isaac Han**, Sangyeon Park, Seungwon Oh, Donghu Kim, Hojoon Lee, Kyung-Joong Kim
- [project page](#) / [paper](#) / [code](#)

Activation by Interval-wise Dropout: A Simple Way to Prevent Neural Networks from Plasticity Loss ICML'25

- Sangyeon Park, **Isaac Han**, Seungwon Oh, Kyung-Joong Kim
- [paper](#)

Deep ensemble learning of tactics to control the main force in a real-time strategy game *Multimedia Tools and Applications (2024)*

- **Isaac Han**, Kyung-Joong Kim
- [paper](#)

Monte Carlo and Temporal Difference Methods in Reinforcement Learning [AI-eXplained]

IEEE Computational Intelligence Magazine
(2023)

- **Isaac Han**, Seungwon Oh, Hoyoun Jung, Insik Chung, Kyung-Joong Kim
- [paper](#)

Seamless-walk: natural and comfortable virtual reality locomotion method with a high-resolution tactile sensor

Virtual Reality
(2023)

- Yunho Choi, Dong-Hyeok Park, Sungha Lee, **Isaac Han**, Ecehan Akan, Hyeon-Chang Jeon, Yiyue Luo, Seungjun Kim, Wojciech Matusik, Daniela Rus, Kyung-Joong Kim
- [paper](#)

A New Open-Source Off-Road Environment for Benchmark Generalization of Autonomous Driving

IEEE Access
(2021)

- **Isaac Han**, Dong-Hyeok Park, Kyung-Joong Kim
- [paper](#) / [code](#) (60+ ★)

Preprints and Workshops

Shared Representation for 3D Pose Estimation, Action Classification, and Progress Prediction from Tactile Signals

Under review

- **Isaac Han**, Seoyoung Lee, Sangyeon Park, Ecehan Akan, Yiyue Luo, Joseph DelPreto, Kyung-Joong Kim
- [paper](#) / [video](#)

Recovering Plasticity of Neural Networks via Soft Weight Rescaling

arXiv

- Seungwon Oh, Sangyeon Park, **Isaac Han**, Kyung-Joong Kim
- [paper](#)

Smart Insole: Predicting 3D human pose from foot pressure

2nd NeurIPS Workshop on Touch Processing

- **Isaac Han**, Seoyoung Lee, Sangyeon Park, Ecehan Akan, Yiyue Luo, Kyung-Joong Kim
- [paper](#)

Academic Service

Reviewer	The ACM conference on Human Factors in Computing Systems (CHI)	2026
Reviewer	IEEE International Conference on Robotics & Automation (ICRA)	2026
Reviewer	IEEE Conference on Games (CoG)	2025
Reviewer	IEEE Access	2023

Technical Skills

Proficient: Python, PyTorch, Tensorflow, Jax

Experienced: Docker, Github